

4855452 INTERNATIONAL RECTIFIER

55C 04903 D

Data Sheet No. PD-2.005C

T-01~13

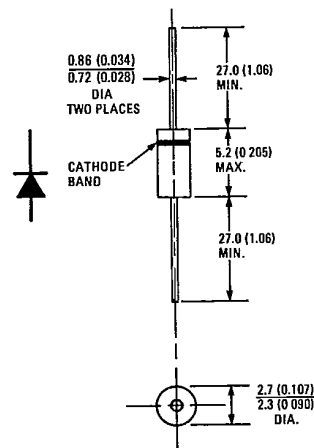
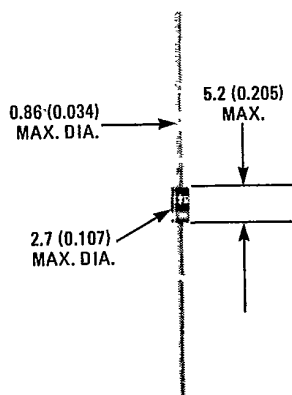
INTERNATIONAL RECTIFIER **1N4001 SERIES****1.0 Amp Silicon Rectifier Diodes****Major Ratings and Characteristics**

	1N4001	Units
$I_{F(AV)}$	1.0 *	A
@ Max. T_A	75 *	$^{\circ}\text{C}$
I_{FSM}		
@ 50 Hz	28.7	A
@ 60 Hz	30 *	A
$I^2 t$	58.1	$\text{A}^2 \sqrt{\text{s}}$
T_J Range	-65 to 175 *	$^{\circ}\text{C}$
VRRM Range	50 to 1000 *	V

Description/Features

- Economical 1 ampere general purpose diode for industrial application
- Molded epoxy DO-204AL case style
- Low forward voltage drop and high surge rating
- Excellent reliability

*JEDEC registered value.

CASE STYLE AND DIMENSIONS

Case Style DO-204AL (DO-41)
Dimensions in Millimeters and (Inches)

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1N4001 Series

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VOLTAGE RATINGS

Part Number	V _{RRM} — Max. Repetitive Peak Reverse Voltage (V)	V _{R(RMS)} — Max. RMS Reverse Voltage (V)	V _R — Max. DC Blocking Voltage (V)
	T = -65 to 175°C	T = -65 to 175°C	T = -65 to 165°C
1N4001	50*	35	50*
1N4002	100*	70	100*
1N4003	200*	140	200*
1N4004	400*	280	400*
1N4005	600*	420	600*
1N4006	800*	560	800*
1N4007	1000*	700	1000*

ELECTRICAL SPECIFICATIONS

		1N4001	Units	Conditions
I _{F(AV)}	Max. average forward current	1.0*	A	Half sine wave conduction
	@ Max. T _A	75*	°C	①, double side cooled.
I _{FSM}	Max. peak one cycle, non-repetitive surge current	28.7	A	Half cycle 50 Hz sine wave or 6 ms rectangular pulse.
		30*		Half cycle 60 Hz sine wave or 5 ms rectangular pulse.
I ² √t	Max. I ² √t for fusing ①	58.1	A ² √s	t = 0.1 to 10 ms, with V _{RRM} following surge = rated V _{RRM} .
V _{FM}	Max. peak forward voltage	1.1*	V	T _A = -65°C to 75°C, I _F = 1 Adc
		1.6*	V	T _A = -65°C to 75°C, I _{F(AV)} = 1A (3.14A peak)
I _R	Max. dc reverse current	10*	μA	T _A = 25°C
		50*	μA	T _A = 100°C
I _{R(AV)}	Max. average reverse current	30*	μA	T _A = 75°C, I _{F(AV)} = 1A, V _{RRM} = rated V _{RRM}

THERMAL-MECHANICAL SPECIFICATIONS

T _J	Max. operating junction temperature range	-65* to 175*	°C	
T _{stg}	Max. storage temperature range	-65* to 200*	°C	
wt	Approximate weight	0.33 (0.012)	g (oz)	
	Case Style	DO-204AL (DO-41)		

① T_L is measured 8.7 mm (0.344 in.) to 9.5 mm (0.375 in.) from device case.① I²t for time t_x = I²√t · √t_x.

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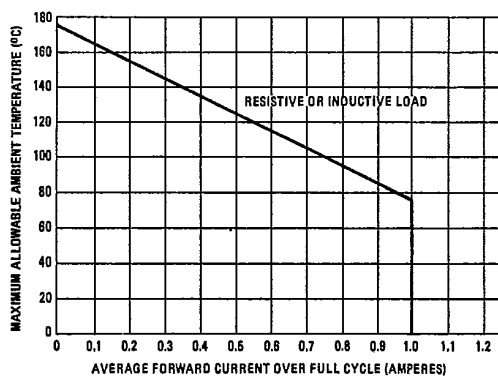


Fig. 1 — Average Forward Current Vs. Maximum Allowable Ambient Temperature

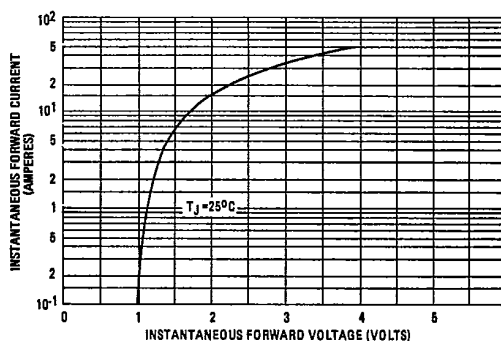


Fig. 2 — Maximum Forward Voltage vs. Forward Current

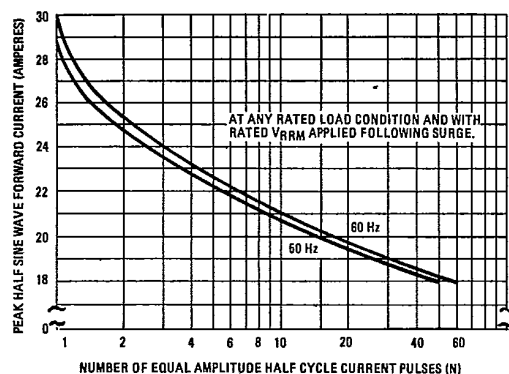


Fig. 3 — Maximum Non-Repetitive Surge Current Vs. Number of Current Pulses

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